PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Success and Challenges of Health System Resilience-enhancing Strategies for Managing Public Health Emergencies of International Concerns (PHEIC): A Systematic Review Protocol
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VERSION 1 – REVIEW

REVIEWER	Sinu Jose Post Graduate Institute of Medical Education and Research, Community medicine and SPH
REVIEW RETURNED	30-Sep-2022

GENERAL COMMENTS	Well presented protocol, significant in the recent pandemic time .

REVIEWER	Bumi Herman Chulalongkorn University, Public Health
REVIEW RETURNED	30-Sep-2022

GENERAL COMMENTS	Thank you for the chance to review an interesting review protocol. I pointed out some issues and am willing to hear any response or rebuttal from the authors. Furthermore, since the review has started, it is imperative to elaborate on its progress of the review. 1. Would the authors explore the health system resilience strategies from the official documents issued by the government of pertinent countries or global organization guidelines? (such as national guidelines, national reports, minister of health decree, etc.). I realized that the authors focus on global studies. However, some national measures are suitable for the global situation. Hence, it is crucial to explore some national or local strategies. 2. Is monkeypox also considered PHEIC? What is the consideration of the author to either include or exclude this recent outbreak? 3. Since the authors have started the review, is there any difference in the search results when adding geographical location as applicable keywords as certain PHEIC began in a specific region? 4. The authors use Joanna Briggs Institute's (JBI) critical appraisal checklist for the qualitative research assessment. Hence, I assume that the paper included in the review is only a qualitative study. However, should the health resilience strategies be written as quantitative research and rely on the quantitative outcome as the evaluation method, how does the author assess this study? Alternatively, how many quantitative studies deem eligible to be included in the review so far? One example of this study is the policy to increase vaccination uptake or reduce transmission by calculating

	the R0 of the disease. 5. It would be great if we could compare the results between countries that applied similar health system resilience strategies as there are no fit-for-all strategies for PHEIC 6. It Seems that most authors come from one country and work for the ministry of a pertinent country. Should the authors find the article from the affiliated country, how would the author reduce the bias from reviewing your "own" policies? 7. The protocol is well written and follows the PRISMA checklist, although no checklist is given as a supplementary file. 8. Meta-analysis might be possible if the health system resilience strategies articles have definite outcomes such as reduction of cases, increased number of prevention programs (such as vaccination), number of drugs procurement, assessment of health burden (health worker burden), or changes of budget allocation for a health emergency. Etc. I would like to know whether some articles focus on the health system resilience and these quantitative impacts.

VERSION 1 – AUTHOR RESPONSE

Reviewer reports
Reviewer 1
1 Well-presented protocol, significant in the recent pandemic time: Thank you

Reviewer 2
1 Would the authors explore the health system resilience strategies from the official documents issued by the government of pertinent countries or global organization guidelines? (Such as national guidelines, national reports, minister of health decree, etc.). I realized that the authors focus on global studies. However, some national measures are suitable for the global situation. Hence, it is crucial to explore some national or local strategies:
We have taken the peer reviewed published articles related to implementation of strategies to improve the quality of the study irrespective of locality. The context is national, regional, and global and all related articles are included in the review.

2 Is monkeypox also considered PHEIC? What is the consideration of the author to either include or exclude this recent outbreak? On July 23, 2022, WHO declared the current monkeypox outbreak a Public Health Emergency of International Concern (PHEIC).
Screening, reference checking of detected studies and hand searching of related journals were conducted from 30th of May to 15th of July 2022. Therefore, it's not included for the previous protocol. However, if investigators would find related journal article regarding monkey pox, we are happy to include those because still data extraction and quality assessment are going on.

3 Since the authors have started the review, is there any difference in the search results when adding geographical location as applicable keywords as certain PHEIC began in a specific region?
Even though the different outbreaks began different regions; we analyse the overall successful strategies for control and challenges encountered. We'll discuss the findings according to the context of the region and give the recommendations.

4 The authors use Joanna Briggs Institute's (JBI) critical appraisal checklist for the qualitative research assessment. Hence, I assume that the paper included in the review is only a qualitative study. However, should the health resilience strategies be written as quantitative research and rely on the quantitative outcome as the evaluation method, how does the author assess this study? Alternatively, how many quantitative studies deem eligible to be included in the review so far? One example of this study is the policy to increase vaccination uptake or reduce transmission by

calculating the R0 of the disease We had to focus on qualitative studies to achieve our objectives. This is a qualitative assessment, and we don't include quantitative analysis.
Agree with the example. However, studies such as prediction modelling based on calculation of Ro are not included as our objective is to identify the success and challenges of already implement health system resilience strategies.

5 It would be great if we could compare the results between countries that applied similar health system resilience strategies as there are no fit-for-all strategies for PHEIC: Thank you for the suggestion. Yes. We are analysing and will compare the countries or regions
6 It Seems that most authors come from one country and work for the ministry of a pertinent country. Should the authors find the article from the affiliated country, how would the author reduce the bias from reviewing your "own" policies?

We didn't include National policies for each and every county. We included peer-reviewed journal articles with implementation of health system resilience enhancing strategies according to WHO Health system 6 Building blocks

7 The protocol is well written and follows the PRISMA checklist, although no checklist is given as a supplementary file: Thank you for the comment. Attached PRISMA checklist as a supplementary file 1

8 Meta-analysis might be possible if the health system resilience strategies articles have definite outcomes such as reduction of cases, increased number of prevention programs (such as vaccination), number of drugs procurement, assessment of health burden (health worker burden), or changes of budget allocation for a health emergency. Etc. I would like to know whether some articles focus on the health system resilience and these quantitative impacts:

We didn't include outcome assessment studies in this review. Therefore, all qualitative studies included for the current review and meta-aggregation will be done for the data synthesis

Thank you for the valuable comments

VERSION 2 – REVIEW

REVIEWER	Bumi Herman Chulalongkorn University, Public Health
REVIEW RETURNED	18-Oct-2022

GENERAL COMMENTS	Thank you for the response from the authors. I am aware that including some health policy studies with quantitative outcomes is not feasible. Looking forward to the published results